

PreflightQC — Instagram / Meta Video Specification Research (Engineering Source Material)

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1. Executive conclusion

PreflightQC can safely and deterministically validate Instagram/Meta video files **only against the Instagram Platform Content Publishing API specifications** (developer documentation), which is the single Tier 1 source that publishes precise, file-measurable, hard container/codec/bitrate/duration rules for Reels and Stories. Everything measurable in those specs — container (MOV/MP4), moov-atom position ("faststart"), edit-list absence, video codec (HEVC/H264), scan type (progressive), GOP closure, 4:2:0 chroma subsampling, frame rate range (23–60 fps), maximum horizontal pixels (1920), video bitrate ceiling (VBR 25 Mbps), audio codec (AAC), audio sample-rate ceiling (48 kHz), audio channel count (1–2), audio bitrate (128 kbps), duration bounds, and file-size ceilings — can be read from a local file with MediaInfo or ffprobe and are appropriate FAIL-safe rules **for the API-publishing use case**. Violating these specs produces a hard API rejection at container creation (a spec violation returns a Graph API error rather than a warning), which is what makes them safe to treat as FAIL for that target.

The **Meta Ads Guide / Business Help Centre** specs (for paid placements) are largely *recommendations* ("recommended", "should"), not hard requirements, and several of the per-placement numeric values are JavaScript-rendered and could not be extracted verbatim from Tier 1 pages during this research — those must be treated as WARN-only or UNVERIFIED, not FAIL. Two categories the brief asked about are **explicitly UNKNOWN at Tier 1**: Meta publishes **no** audio loudness normalisation target (no LUFS/LKFS/dBFS true-peak figure) in any Tier 1 documentation, and Meta publishes **no** creator-facing HDR/colour-space requirement for uploads (the only Tier 1 HDR material is an engineering blog describing server-side tone-mapping, not an input spec). Do not import broadcast or third-party LUFS numbers.

Bottom line: build the validation engine around the Content Publishing API spec as the authoritative FAIL set for organic Reels/Stories, treat all Ads Guide placement numbers as WARN/informational, and mark loudness and HDR as UNKNOWN/informational.

2. Official source register

S1 — Content Publishing (Instagram Platform), Meta for Developers. URL:

<https://developers.facebook.com/docs/instagram-platform/content-publishing/> — Title: "Content Publishing - Meta for Developers". Accessed Aug 2026. Visible versioning: API v25.0; dated updates (alt_text 24 Mar 2025; user_tags for stories 9 Jul 2025).  CONFIDENCE: HIGH.

S2 — IG User Media reference (Reel/Story video specifications), Meta for Developers. URL: <https://developers.facebook.com/docs/instagram-platform/instagram-graph-api/reference/ig-user/media/> — Title: "IG User Media". Accessed Aug 2026. Contains the explicit "Reel Specifications" and "Story Video Specifications" blocks, plus the Reels-tab eligibility statement. CONFIDENCE: HIGH.

S3 — Upload a Video, Meta Business Help Center. URL: <https://www.facebook.com/business/help/163131210928551> — Title: "Upload a Video | Meta Business Help Center". Accessed Aug 2026. States "Videos must be less than 240 minutes long", "Stereo AAC audio compression with 128kbps or more is recommended", "MP4 format is recommended but nearly all types of video files are supported". Facebook CONFIDENCE: HIGH (wording), MEDIUM (currency — page still references Creator Studio).

S4 — Specifications for Live Video on Facebook, Meta Business Help Center. URL: <https://www.facebook.com/business/help/162540111070395> — Title: "Specifications for Live Video on Facebook". Accessed Aug 2026. RTMPS/live only; audio sample rate 44.1 kHz or 48 kHz; H.264 Level 4.1/4.2; keyframe 2 s recommended, 4 s max; 8-hour max. CONFIDENCE: HIGH but scope = live streaming, not file upload.

S5 — Ads Guide, Instagram Feed video (Awareness). URL: <https://www.facebook.com/business/ads-guide/update/video/instagram-feed> — Title: "Awareness Video Ad Specs on Instagram Feed". Accessed Aug 2026. JavaScript-rendered; values captured via indexed snippet. CONFIDENCE: MEDIUM.

S6 — Ads Guide, Instagram Stories video (Awareness). URL: <https://www.facebook.com/business/ads-guide/update/video/instagram-story> — Title: "Awareness Video Ad Specs on Instagram Stories". Accessed Aug 2026. Body partially JS-rendered; duration/file-size/min-width and card-split behaviour captured. CONFIDENCE: MEDIUM.

S7 — Ads Guide, Instagram Reels video (Awareness). URL: <https://www.facebook.com/business/ads-guide/update/video/instagram-reels> — Title: "Awareness Video Ad Specs on Instagram Reels". Accessed Aug 2026. Body NOT extractable (returned placeholder). CONFIDENCE: LOW / BLOCKED.

S8 — Ads Guide, Facebook Feed video (Awareness). URL: <https://www.facebook.com/business/ads-guide/update/video> — Title: "Awareness Video Ad Specs on Facebook Feed". Accessed Aug 2026. Verbatim "H.264 compression, square pixels, fixed frame rate, progressive scan and stereo AAC audio compression at 128kbps+"; "Videos should not contain edit lists or special boxes in file containers". CONFIDENCE: HIGH (wording).

S9 — Instagram Help Center, Reels length. URL: <https://help.instagram.com/2720958398006062> — Title: "Help Center" (meta-description "Record and edit videos up to 3 minutes with Instagram Reels"). Instagram Accessed Aug 2026. CONFIDENCE: HIGH (for the 3-minute in-app editing statement).

S10 — About Instagram, Reels feature page. URL: <https://about.instagram.com/features/reels> — Title: "Instagram Reels | Share & Create Short Videos". Accessed Aug 2026. "Create multi-clip videos up to 3 minutes"; "Reels over 3 minutes won't be recommended to new audiences". Instagram

CONFIDENCE: HIGH. (Context: the 3-minute upload limit was announced by Instagram head Adam Mosseri in January 2025 — “Starting today, you can upload Reels to Instagram up to three minutes long... historically, it’s only been 90 seconds” Social Media Today — doubling the prior 90-second cap.)

S11 — Engineering at Meta, “Bringing HDR video to Reels”. URL:

<https://engineering.fb.com/2023/07/17/video-engineering/hdr-video-reels-meta/> — Published 17 Jul 2023, authored by Adarsh Solanki, Cosmin Stejerean and Chris Ellsworth. Describes server-side HDR ingest + tone-mapping to BT.709 SDR; states Meta sees “millions of HDR videos uploaded every day” fb and renders SDR UI overlays over HDR video rather than imposing an HDR input spec.

CONFIDENCE: HIGH (as evidence that no input HDR *requirement* exists).

S12 — Video API Overview / Publishing, Meta for Developers. URLs:

<https://developers.facebook.com/docs/video-api/overview/> and <https://developers.facebook.com/docs/video-api/guides/publishing/> — Accessed Aug 2026.

Resumable Upload API; file_type valid values include video/mp4. CONFIDENCE: HIGH.

3. Instagram Reels rule table (primary source: S2 Content Publishing API)

All rows below are verbatim from S2 unless noted. RULE TYPE for the API-publishing use case = HARD REQUIREMENT (the API rejects containers that violate them); for organic in-app posting, treat as DOCUMENTED LIMIT.

RULE	VALUE	SOURCE	CONFIDENCE	RULE TYPE	Offline-measurable?
Container	MOV or MP4 (MPEG-4 Part 14)	S2	HIGH	HARD (API)	Yes (major_brand / container)
Edit lists	None (“no edit lists”)	S2	HIGH	HARD (API)	Yes (elst atom)
moov atom position	At front of file (faststart)	S2	HIGH	HARD (API)	Yes (atom order)
Video codec	HEVC or H264	S2	HIGH	HARD (API)	Yes
Scan type	Progressive	S2	HIGH	HARD (API)	Yes (scan_type)
GOP	Closed GOP	S2	HIGH	HARD (API)	Partially (derivable)
Chroma	4:2:0	S2	HIGH	HARD (API)	Yes (pix_fmt)

subsampling					
Frame rate	23–60 FPS	S2	HIGH	HARD (API)	Yes
Max horizontal pixels	1920 columns	S2	HIGH	DOCUMENTED LIMIT	Yes (width)
Aspect ratio	Between 0.01:1 and 10:1; 9:16 recommended	S2	HIGH	HARD range + RECOMMENDATION	Yes
Video bitrate	VBR, 25 Mbps maximum	S2	HIGH	DOCUMENTED LIMIT	Yes
Audio codec	AAC	S2	HIGH	HARD (API)	Yes
Audio sample rate	48 kHz maximum	S2	HIGH	DOCUMENTED LIMIT	Yes
Audio channels	1 or 2 (mono/stereo)	S2	HIGH	HARD (API)	Yes
Audio bitrate	128 kbps	S2	HIGH	RECOMMENDATION/LIMIT	Yes
Duration	15 minutes maximum, 3 seconds minimum	S2	HIGH	HARD (API)	Yes
File size	300 MB maximum	S2	HIGH	HARD (API)	Yes

Reels-tab eligibility (S2, verbatim): “Although you can publish reels with longer durations and different aspect ratios, only reels with durations between 5 and 90 seconds and aspect ratios of 9:16 are eligible to appear in the Reels tab.” Medium This is an *eligibility* condition, not an upload requirement — treat any 5–90 s / 9:16 check as WARN/INFO, never FAIL.

Conflict flag (Reels duration/size): S2 (API) says **15 min max / 300 MB max**. The in-app experience (S9/S10) says Reels can be created “up to 3 minutes” in the editor (announced Jan 2025). Metricool Ads-placement Reels values (S7) were not extractable. Note also that some third-party integrator docs (e.g. ContentStudio) state a **100 MB** Reels ceiling, Contentstudio conflicting with the Tier 1 S2 figure of **300 MB** — trust S2. These are different surfaces with different limits — see §10.

4. Feed rule table

Instagram **Feed video** has **no dedicated file-spec block in the Content Publishing API** (the API's video path inherits the general VIDEO container; the explicit spec blocks in S2 are only for Reels and Stories). Feed video file specs therefore come from the Ads Guide (S5) and general upload help (S3), which are mostly recommendations.

RULE	VALUE	SOURCE	CONFIDENCE	RULE TYPE	Offline-measurable?
File type	MP4, MOV or GIF	S5	MEDIUM	RECOMMENDATION	Yes
Aspect ratio	9:16 (Awareness variant); tolerance 1%	S5	MEDIUM	RECOMMENDATION	Yes
Resolution	1080 × 1920 px	S5	MEDIUM	RECOMMENDATION	Yes
Minimum width	250 px	S5	MEDIUM	DOCUMENTED LIMIT	Yes
Max file size	4 GB	S5	MEDIUM	DOCUMENTED LIMIT	Yes
Duration	1 second to 60 minutes	S5	MEDIUM	DOCUMENTED LIMIT	Yes
Captions / Sound	Optional, but recommended	S5	MEDIUM	RECOMMENDATION	N/A / partially
Edit lists / special boxes	Should not contain	S5/S8	MEDIUM	RECOMMENDATION	Yes

Conflict flag (Feed aspect ratio): the Tier 1 “Awareness” snippet lists 9:16 / 1080×1920 / min width 250 px, while other objective pages and Tier 2 transcriptions of the same Ads Guide list 4:5 / “at least 1080×1080” / min width 500 px. Meta appears to differ by objective and to have shifted Feed toward 9:16. Document both; do not FAIL on either — see §10.

5. Stories rule table (primary source: S2 Content Publishing API)

RULE	VALUE	SOURCE	CONFIDENCE	RULE TYPE	Offline-measurable?
Container	MOV or MP4 (MPEG-4 Part 14), no edit lists, moov at	S2	HIGH	HARD (API)	Yes

	front				
Video codec	HEVC or H264, progressive, closed GOP, 4:2:0	S2	HIGH	HARD (API)	Yes
Frame rate	23–60 FPS	S2	HIGH	HARD (API)	Yes
Max horizontal pixels	1920 columns	S2	HIGH	DOCUMENTED LIMIT	Yes
Aspect ratio	Between 0.1:1 and 10:1; 9:16 recommended	S2	HIGH	HARD range + RECOMMENDATION	Yes
Video bitrate	VBR, 25 Mbps maximum	S2	HIGH	DOCUMENTED LIMIT	Yes
Audio codec	AAC, 48 kHz max, 1–2 channels	S2	HIGH	HARD (API)	Yes
Audio bitrate	128 kbps	S2	HIGH	RECOMMENDATION/LIMIT	Yes
Duration	60 seconds maximum, 3 seconds minimum	S2	HIGH	HARD (API)	Yes
File size	100 MB maximum	S2	HIGH	HARD (API)	Yes

Note (Stories aspect-ratio lower bound): S2 lists the Story lower bound as **0.1:1** while the Reels lower bound is **0.01:1** [facebook](#) — a genuine one-order-of-magnitude difference in the same document. Preserve both values exactly; do not normalise. Ads Guide Stories (S6) confirms duration “1 second to 60 minutes”, file size 4 GB, min width 250 px for the *ads* surface, plus card-split behaviour (ads <35 s play in full; ads ≥15 s may split into 1–3 Stories cards) — a conflict with the API’s 60-second / 100 MB organic-story limits (§10).

6. Relevant Facebook/Meta rules (same validation engine)

- **Facebook Feed video ad settings (S8, verbatim):** “H.264 compression, square pixels, fixed frame rate, progressive scan and stereo AAC audio compression at 128kbps+”; “Videos should

not contain edit lists or special boxes in file containers." Facebook RULE TYPE: RECOMMENDATION. All offline-measurable, including "fixed frame rate" (CFR vs VFR).

- **General video upload (S3):** "Videos must be less than 240 minutes long"; "Stereo AAC audio compression with 128kbps or more is recommended"; "MP4 format is recommended but nearly all types of video files are supported." Facebook RULE TYPE: RECOMMENDATION / DOCUMENTED LIMIT (240 min).
- **Resumable Upload / Video API (S12):** file_type valid values include video/mp4 ; single-request vs resumable upload thresholds are server/transport limits, not file-content specs.
- **Facebook Live (S4):** audio sample rate 44.1 kHz or 48 kHz; H.264 Level 4.1 ($\leq 1080p30$) / Level 4.2 (1080p60); keyframe interval 2 s recommended, 4 s max; 8-hour max. Facebook Scope = live streaming only; include as informational, not as a file-upload FAIL set.

7. FAIL rules (safe to reject a file)

Apply as FAIL **only when the user has selected the API-publishing target (organic Reels/Stories)**, since these are the only Tier 1 hard, file-measurable requirements (all from S2), and violating them produces a hard Graph API rejection:

1. Container is not MOV/MP4 (MPEG-4 Part 14).
2. Video codec is not HEVC or H264.
3. Chroma subsampling is not 4:2:0.
4. Scan type is not progressive.
5. Audio codec present but not AAC.
6. Audio channels > 2.
7. Reels: duration < 3 s or > 15 min; Stories: duration < 3 s or > 60 s.
8. Reels: file size > 300 MB; Stories: file size > 100 MB.
9. Frame rate outside 23–60 fps.
10. Aspect ratio outside 0.01:1–10:1 (Reels) / 0.1:1–10:1 (Stories).
11. Horizontal pixels > 1920.
12. Video bitrate > 25 Mbps.
13. Audio sample rate > 48 kHz.
14. Edit lists present, or moov atom not at front (faststart absent).

Everything else — including all Ads Guide placement numbers and the 5–90 s / 9:16 Reels-tab eligibility window — must NOT be a FAIL.

8. WARN rules (flag, do not reject)

- Aspect ratio not 9:16 (recommended, not required) for Reels/Stories.
- Reels duration outside the 5–90 s Reels-tab eligibility window (eligibility, not a hard limit).
- Feed aspect ratio not 4:5 / 9:16; below 1080 px width; landscape 16:9.
- Audio bitrate below 128 kbps (S2/S3 phrase it as recommended).
- Frame rate not “fixed” (VFR detected) — S8 recommends fixed frame rate.
- Edit lists / special boxes present when target is Ads/Feed (S8 says “should not”).
- File approaching the 4 GB Ads ceiling, or a Stories file above ~200 MB for reliability.
- Resolution below 1080p (recommended baseline in help/ads material).

9. Informational rules (metadata to display, never gate)

- Overall duration, per-stream durations, container major_brand, faststart status.
- Colour primaries / transfer characteristics / matrix coefficients, bit depth, HDR metadata (HDR10/HLG/Dolby Vision) — display only; see §10 HDR UNKNOWN.
- GOP length / keyframe interval, pixel aspect ratio, stream counts, per-stream and overall bitrate, audio sample rate, channel layout.
- Integrated loudness / true peak (LUFS/dBTP) if the engine measures them — display only, since Meta specifies no target (see §10).

10. Conflicting / uncertain specifications

1. **Reels duration:** API (S2) = 15 min max; in-app editor (S9/S10, Mosseri Jan 2025) = 3 min; some 2026 third-party claims of 20-min uploads are NOT Tier 1. Do not FAIL beyond the API’s 15-min for the API use case; mark others informational.
2. **Reels/Stories vs Ads durations & sizes:** API organic Stories = 60 s / 100 MB; Ads Guide Stories (S6) = 1 s–60 min / 4 GB. Different products (organic publish vs paid placement). Keep separate rule sets per target.
3. **Feed aspect ratio (S5):** 9:16 @1080×1920 (Awareness snippet) vs 4:5 @≥1080×1080 (other objectives / Tier-2 transcription). Documented conflict; WARN only.
4. **Aspect-ratio lower bound within S2 itself:** Reels 0.01:1 vs Stories 0.1:1. Preserve verbatim.
5. **Reels file size:** S2 (Tier 1) = 300 MB max; some integrator docs (ContentStudio) cite 100 MB. Trust S2’s 300 MB; flag the discrepancy.
6. **Audio sample rate:** S2 = 48 kHz max (Reels/Stories); S4 (Live) = 44.1 or 48 kHz. Scope differs.

7. **Reels ad video specs (S7):** Tier 1 page body was not extractable during research → BLOCKED/UNVERIFIED. Third-party transcriptions (H.264, 9:16, 500×888 min, 0 s–15 min, 4 GB) exist but are not Tier 1; do not treat as authoritative.
8. **Audio loudness (LUFS/LKFS/dBFS true peak): UNKNOWN at Tier 1.** No Meta Business Help Centre, Ads Guide, or developer page located in this research states any loudness normalisation target. The –14 LUFS figure people cite is a music-streaming norm (Spotify, YouTube, TIDAL, Amazon Music and SoundCloud normalise to –14 LUFS integrated; Deezer –15; Apple Music –16) Fora Soft — none of which are Meta/Instagram targets. Mark UNKNOWN; do NOT import.
9. **HDR / colour space: UNKNOWN at Tier 1 as an input requirement.** Meta supports HDR uploads to Reels and tone-maps server-side fb to BT.709 SDR (S11, the “Bringing HDR video to Reels” engineering blog by Solanki, Stejerean and Ellsworth), but publishes no creator-facing HDR/colour-space *requirement* (no HDR10/HLG/Rec.2020/Rec.709 mandate) for Instagram/Facebook uploads. The Content Publishing API specs (S2) are silent on colour space for video (they specify sRGB only for still images/cover photos). facebook Mark HDR/colour space as informational/UNKNOWN.

11. Metadata fields required from inspection engine (MedialInfo/ffprobe)

Container/format: `format_name`, `major_brand` / `compatible_brands`, `format_long_name`, `faststart` / moov-atom position, presence of edit list (`elst`), overall bitrate, file size, overall duration, stream count. Video stream: `codec_name` (h264/hevc), `profile`, `level`, `width`, `height`, display aspect ratio, sample/pixel aspect ratio, `pix_fmt` (chroma subsampling + bit depth), `field_order` / scan type, `r_frame_rate` / `avg_frame_rate`, CFR vs VFR, `bit_rate`, GOP/keyframe interval (derivable from packet/keyframe analysis), `color_primaries`, `color_transfer`, `color_space` / matrix, HDR side-data / mastering-display metadata. Audio stream: `codec_name` (aac), `sample_rate`, `channels` / channel layout, `bit_rate`. Optional loudness pass: integrated LUFS + true peak (display only).

12. Proposed machine-readable rule structure

A rule set keyed by **target** (`ig_reels_api`, `ig_stories_api`, `ig_feed`, `fb_feed`, `meta_ad_*`) where each rule carries: `id`, `parameter` (mapped to an ffprobe/MedialInfo field), `operator` (`in`, `range`, `eq`, `lte`, `gte`, `present`, `absent`), `value` (verbatim from source, units explicit), `severity` (`FAIL` / `WARN` / `INFO`), `rule_type` (`HARD` / `RECOMMENDATION` / `DOCUMENTED_LIMIT` / `UNKNOWN`), `source_id` (S1–S12), `source_url`, `confidence`, and `measurable_offline` (bool). Conflicting values are stored as an array under the same `parameter` with per-entry `source_id`, so the engine surfaces both rather than silently choosing. The 5–90 s / 9:16 Reels-tab eligibility condition is stored as a separate eligibility rule at `severity: WARN`. Loudness and HDR entries exist with `severity: INFO`, `rule_type: UNKNOWN`, and an explicit `no_official_target: true` flag.

13. Engineering caveats

- The only precise, hard, file-level Instagram video spec Meta publishes is the **Content Publishing API** block (S2) for Reels and Stories; treat it as the FAIL backbone but scope it to the API-publishing use case, since organic in-app posting and paid placements tolerate more.
- Ads Guide placement pages are JavaScript-rendered and partially non-extractable by automated tooling; the Reels ad video page (S7) yielded no body. Any numbers taken from Tier-2 transcriptions must be labelled MEDIUM/LOW and never used as FAIL.
- "Closed GOP", "no edit lists", and "moov atom at front" are all real ffprobe/MediaInfo-measurable properties but require packet-level or atom-level inspection, not just top-level metadata.
- Reels-tab eligibility (5–90 s, 9:16) is a discovery condition, not an acceptance requirement — never FAIL on it.
- Server-side transcoding outcomes, account eligibility, remixing/branded-content flags, text/logo safe zones, and algorithmic reach are **not** offline-measurable and must never be validation rules.
- Specs change without notice; store `source_url + date_accessed` per rule and re-verify before release.
- Do not import broadcast-standard or music-streaming LUFS values (e.g. -14 LUFS) or HDR numbers to fill the loudness/HDR gaps; leave them UNKNOWN.

14. SOURCE PACK STATUS: PARTIAL

Tier 1 coverage is COMPLETE and HIGH-confidence for the Content Publishing API (Reels + Stories file specs, and Reels-tab eligibility), general upload help, Facebook Feed ad video settings, Instagram Reels length, and the HDR engineering context. It is PARTIAL/BLOCKED for the per-objective Instagram Ads Guide numeric tables (JavaScript-rendered; Reels ad video page not extractable), and confirms two deliberate Tier 1 gaps as UNKNOWN: audio loudness targets and HDR/colour-space input requirements.